

JOCELYN LI

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RESEARCH INTERESTS

My research interests broadly lie in computer systems architecture, with a focus on domain-specific accelerator design, GPU/CPU microarchitecture, and developing frameworks that streamline architectural design.

EDUCATION

Princeton University

Princeton, NJ

Bachelor's of Science and Engineering in Computer Science. Minor in Statistics & Machine Learning.

May 2026

- **Unweighted GPA:** 3.95
- **Senior Thesis:** *Improving Speculative Window Decoder Performance for Quantum Error Correction Under Realistic Circuit Conditions* (Advisor: Professor Margaret Martonosi)

RESEARCH EXPERIENCE

Martonosi Research Group

Princeton, NJ

Undergraduate Researcher, PI: Professor Margaret Martonosi

Improving Speculative Window Decoders for Quantum Error Correction

September 2025 – Present

- **Improving speculative window decoder performance** for quantum error correction by **designing adaptive speculation strategies** that improve decoder throughput and reduce wasted computational resources.
- Developing adaptive strategies that adjust speculation based on circuit context and decoder confidence metrics to improve speculation accuracy and minimize misprediction penalty overhead.
- Extended the decoder simulator to incorporate circuit-level variability and realistic noise models, enabling analysis of decoder performance under realistic circuit conditions and supporting the development of adaptive speculation strategies.

Statistical Assertions for Debugging Quantum Circuits and States in CUDA-Q

January – May 2025

- Introduced a **statistical assertion-based debugging workflow** for **CUDA-Q**, enabling validation of qubit states at any point within a quantum circuit, and providing more accessible, efficient, and rigorous debugging for CUDA-Q programs.
- Improved product-state assertion accuracy under sparse measurement distributions by using Fisher's Exact Test and the Monte Carlo Method, making the tool more robust and reliable.
- Integrated the workflow into CUDA-Q's kernel-based framework via a custom **StatKernel class**, which extends CUDA-Q's base kernel class and adds classical, uniform, and product state assertion functions as instance methods.
- Tested the debugging workflow on a range of quantum circuits, including custom circuits designed with deliberate bugs, confirming that well-placed assertions can pinpoint the location and nature of bugs.

Performance Characterization of Quantum Simulation in CUDA-Q

January – May 2025

- Contributed to project conducting **systems-level performance characterization** of NVIDIA's CUDA-Q package, benchmarking Python vs C++ overhead and A100 vs V100 GPU performance using Nsight profiling.
- Assisted with environment setup and execution of large job workflows involving quantum circuits with up to 32 qubits.
- Observed minimal Python overhead and generally higher A100 performance. Deeper circuits showed slower A100 results due to increased computational complexity, stalls from dependency chains, and kernel implementation differences.

INDUSTRY EXPERIENCE

NVIDIA – Hardware Infrastructure Organization

Santa Clara, CA

Intern on GPU Modeling Team

Summer 2025

- Developed a **pipeline visualization tool** for the frontend of NVIDIA's core **GPU architectural model**, supplementing traditional ASCII logs and replacing most manual log inspection with a more efficient, clear interface for debugging one of the model's most complex, error-prone stages.
- Designed an optimized **SQLite database** to store the model's data (up to 80 GB). Achieved real-time data retrieval and improved efficiency in storage and query speed through **indexed JOIN queries** and **schema redesign**.
- Built a **Curses-based TUI** to visualize real-time GPU frontend pipeline object flow, using filters, color-coded stall indicators, and navigation tools (bookmarks, regex search) to improve observability and usability for massive streams of data, enhancing overall debugging efficiency.

- Key contributor in a team that developed prototype architecture for a long-standing distributed test automation framework, which is responsible for more than 50% of traffic at LSF NVIDIA HPC clusters.
- Designed a configuration launcher, which included a form with dropdown menus, dynamically changing tables, selects, and more using **React**, **Vite**, **CSS**, **Axios**, and an internal UI library.
- Used **FastAPI** to design **RESTful APIs** for test configuration processing, with updates reflected on the frontend in real time. Verified backend logic using **PyTest** unit tests.
- Leveraged **Redis Streaming** for log storage and retrieval. Used **websockets** for real-time log transmission and system command/control. Built an access-controlled configuration storage & retrieval system using **MongoDB** and **Azure** roles.

U.S. Environmental Protection Agency - Office of Enterprise Information Programs

Washington D.C.

Intern on Records Management Technology Team

Summer 2023

- Developed self-service tool allowing users to download thousands of records at a time from EPA's Records Management System, automating a previously manual workflow. Used **Boto3** to manage those records in **AWS S3 Bucket**.
- Utilized **SQLite** and **Pandas** to parse JSON metadata and organize data into search result and Excel tables for requestors.
- Notified users via email about recently chunked and zipped search results records that were uploaded to the S3 bucket automatically using **Microsoft Graph API**.

PUBLICATIONS AND PRESENTATIONS

- **J. Li**, E. Rubinshtein, M. Martonosi. *Statistical Assertions for Debugging Quantum Circuits and States in CUDA-Q*. In Proceedings of the 2025 IEEE International Conference on Quantum Computing and Engineering (QCE). Presented at QCE 2025. [[Paper](#)] [[Code](#)]
- E. Rubinshtein, **J. Li**, and M. Martonosi. *Performance Characterization of Quantum Simulation in CUDA-Q*. Poster presented at 2025 IEEE International Conference on Quantum Computing and Engineering (QCE). [[Poster PDF](#)] [[Paper](#)]

HONORS AND AWARDS

- Exemplary Spring 2025 CS Independent Work Award (2025)
Awarded to 5 students out of all 150+ CS Independent Work students.
- Princeton CS department's nominee for Computing Research Association undergraduate research award (2025)
Awarded to 3 students out of all CS students.
- Tau Beta Pi Member (2025)
Awarded to the top 1/5 of the senior engineering class and the top 1/8 of the junior engineering class.
- Manfred Pyka Memorial Physics Prize (2023)
Awarded to outstanding Physics undergraduates who have shown excellence in course work and promise in independent research.

TEACHING EXPERIENCE**Princeton University**

Princeton, NJ

Computer Architecture (COS/ECE 375) Teaching Assistant

Fall 2025

- Hosted office hours to support students with questions on course concepts, assignments, and project work.
- Created exam questions and assisted with grading assignments and exams.

Introduction to Systems / Data Structures and Algorithms (COS 217 / 226) Teaching Assistant

Fall 2024

- Hosted office hours to help students debug their programming projects and assignments.

ACADEMIC PROJECTS

- Designed and implemented an out-of-order superscalar processor in Verilog, and evaluated the performance impact of multiple branch predictor designs on it. (Spring 2025)
- Extend CIRCT to support memory constructs during BTOR2 lowering and add nondeterministic modeling of undefined memory behavior, enabling formal verification of hardware designs with realistic memory semantics. (Fall 2025)

SKILLS

- Programming Languages: Java, Python, C, C++, Assembly, Verilog, Q#, Qiskit, CUDA-Q, R, HTML, JavaScript, CSS
- Tools: Linux, SQL, Emacs, NumPy, Matplotlib, Pandas, Git, PyTorch, MongoDB, Node.js, Flask, AWS